2.1 Defining your objectives

One of the most common mistakes when building a digital twin is to start with the technology rather than the objectives or strategic goals. You may naturally get excited about creating high fidelity 3D visualizations, or building and deploying sensors before answering the fundamental question of 'what business problem are we trying to solve?'. This failure can lead to wastage and digital twins that do not deliver on the promised value.

The success of any digital twin initiative hinges on clearly defining your objectives before beginning the complex process of capturing and modeling physical systems. This objective-first approach serves as the foundation for all subsequent technical decisions, resource allocation, and validation criteria throughout the digital twin lifecycle. By aligning technical capabilities with business imperatives, you can avoid the common trap of building sophisticated digital models that fail to address real operational challenges or deliver measurable value.

2.1.1 Identify specific challenges and opportunities

Start by documenting concrete business problems that impact your operations rather than leading with available technologies. Imagine a plant superintendent facing 12 hours of unplanned downtime per month from centrifugal pump bearing failures, costing $15,000 per hour in lost production, (or $2.16 million annually). Current time-based, or periodic, maintenance replaces bearings every 6 months, meaning that:

- 40% of the bearings are replaced while they are still in good condition, costing $48,000.
- 15% of bearings fail before the scheduled replacement (causing the downtime).
- When a bearing fails, it can take up to 2 weeks to procure a new one from suppliers, so spare parts must be held in inventory.

The opportunity is to switch from planned to predictive maintenance, servicing equipment based on its actual health. Identifying potential issues